```
#!/bin/bash

db="max30102.db"
table="heart_rate_data"

echo "—— WELCOME ——"

while [ 1 ]
do
        echo "Please select any query by typing its sr. no."

        echo " "
        echo "1. View heart rate data (last 50 entries) of… "
        echo "2. Calculate average heart rate of… "
        echo "3. Compare average heart of users."
        echo "4. View oxygen saturation (SpO2) data (last 50 entries) of… "
        echo "5. Calculate average SpO2 of… "
        echo "6. Compare average SpO2 of all users."

        echo " "
        echo -n "Enter sr. no. of query: "
        read choice

        if [ "$choice" = "" ];then echo "Value needed"; exit 1; fi

        case $choice in
                1)
                        echo " "
                        echo "Select (complete the sentence):"
                        echo "    1.  …a person (enter name)."
                        echo "    2.  …all users."
                        echo " "
                        echo -n "Enter your choice: "
                        read ans

                        if [ "$ans" = 1 ]; then
                                echo -n "Enter person name: "
                                read person

                                echo " "
                                sqlite3 -header -column "$db" "SELECT timestamp,
bpm FROM $table WHERE person='$person' ORDER BY id DESC LIMIT 50;"

                        else
                                echo "Displaying heart rate for all users."
                                echo " "
                                sqlite3 -header -column "$db" "SELECT timestamp,
bpm FROM $table ORDER BY id DESC LIMIT 50;"
```

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                        fi
                        ;;

                2)
                        echo " "
                        echo -n "Calculate average HR for: "
                        read person

                        echo " "
                        sqlite3 -header -column "$db" "SELECT AVG(bpm) FROM $table
WHERE person='$person';"
                        ;;

                3)
                        echo " "
                        echo "Comparing average heart rate of all users:"

                        echo " "
                        sqlite3 -header -column "$db" "SELECT person, AVG(bpm) FROM
$table GROUP BY person;"
                        ;;

                4)
                        echo " "
                        echo "Select (complete the sentence):"
                        echo "    1.  ... a person (enter name)."
                        echo "    2.  ... all users."
                        echo " "
                        echo -n "Enter your choice: "
                        read ans

                        if [ "$ans" = 1 ]; then
                                echo -n "Enter person name: "
                                read person

                                echo " "
                                sqlite3 -header -column "$db" "SELECT timestamp,
spo2 FROM $table WHERE person='$person' ORDER BY id DESC LIMIT 50;"

                        else
                                echo "Displaying SpO2 data for all users."

                                echo " "
                                sqlite3 -header -column "$db" "SELECT timestamp,
person, spo2 FROM $table ORDER BY id DESC LIMIT 50;"

                        fi
                        ;;
```

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                5)
                        echo " "
                        echo -n "Enter name of person: "
                        read person

                        echo " "
                        sqlite3 -header -column "$db" "SELECT AVG(spo2) as
Average_SpO2 FROM $table WHERE person='$person';"
                        ;;

                6)
                        echo " "
                        echo "Comparing average SpO2 of all users:"

                        echo " "
                        sqlite3 -header -column "$db" "SELECT person, AVG(spo2) as
Average_SpO2 FROM $table GROUP BY person;"
                        ;;

                7)
                        echo " "
                        echo -n "Enter name of person: "
                        read person

                        echo " "
                        sqlite3 -header -column "$db" "SELECT MAX(bpm) AS max_bpm,
MIN(bpm) AS min_bpm, AVG(bpm) AS avg_bpm, MAX(spo2) AS max_spo2, MIN(spo2) AS
min_spo2, avg(spo2) AS avg_spo2 FROM $table WHERE person='$person'"
                        ;;

                *)
                        echo
                        echo "Invalid selection. Please enter a number between 1
and 6."
                        ;;

        esac

        echo " "
        echo -n "Do you want to continue? (Y/n): "
        read t_choice

        if [ "$t_choice" = 'n' ]; then echo "Exiting program..."; exit 0;
        else clear; fi;

done
```
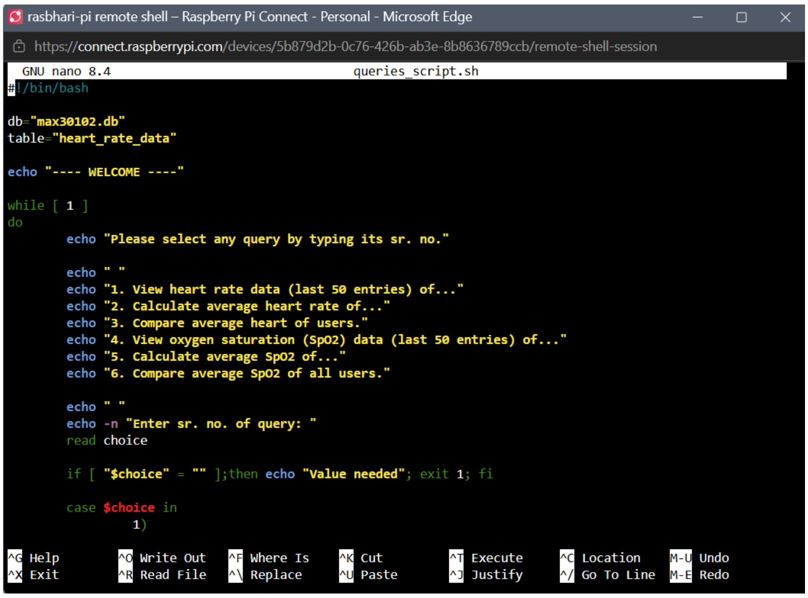
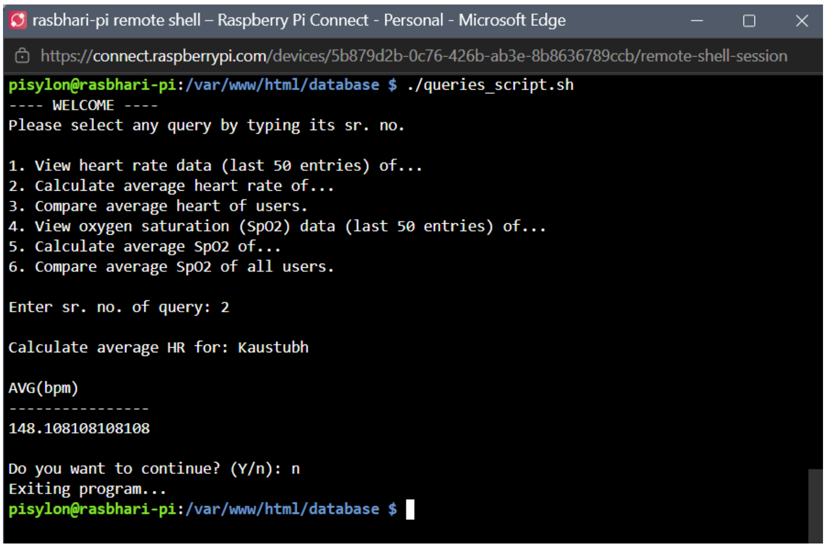



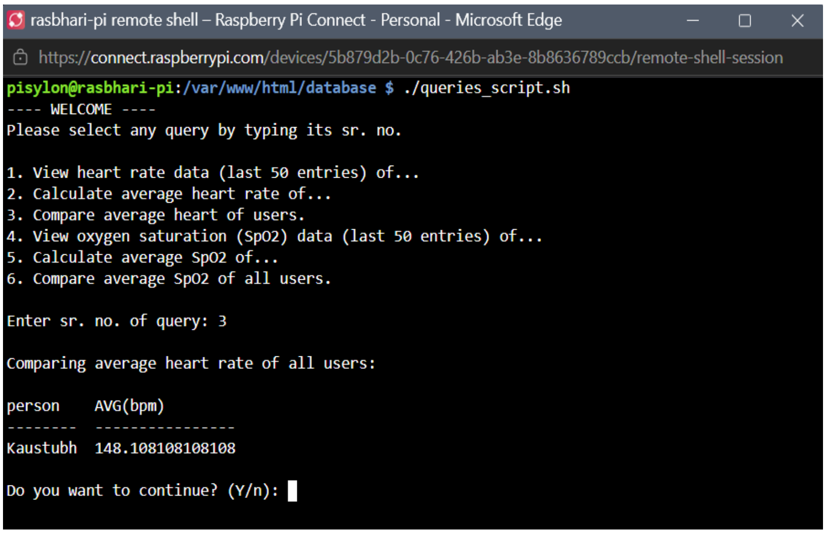
This code is written in Nano text editor whilst connected to Raspberry-Pi Model 2B via Raspberry-Pi Connect.

SSH was also possible but had to keep switching from terminal to screen (of RPi) because of Node-Red.

This is one of the queries being executed. The task is run by typing ‘./<file_name>.sh’ in the command line (Terminal).

It isn’t always possible to run the script after writing it, first it has to be an executable. That was done by the command ‘chmod +x <file_name>.sh’ or ‘chmod 777 <file_name>.sh’.





More people can be added to this database. In-fact, the Node-Red utility is ready to implement more than 1 person, but there was shortage of hardware like ESP-8266 (due to other projects), so that couldn’t be implemented in the given timeframe.